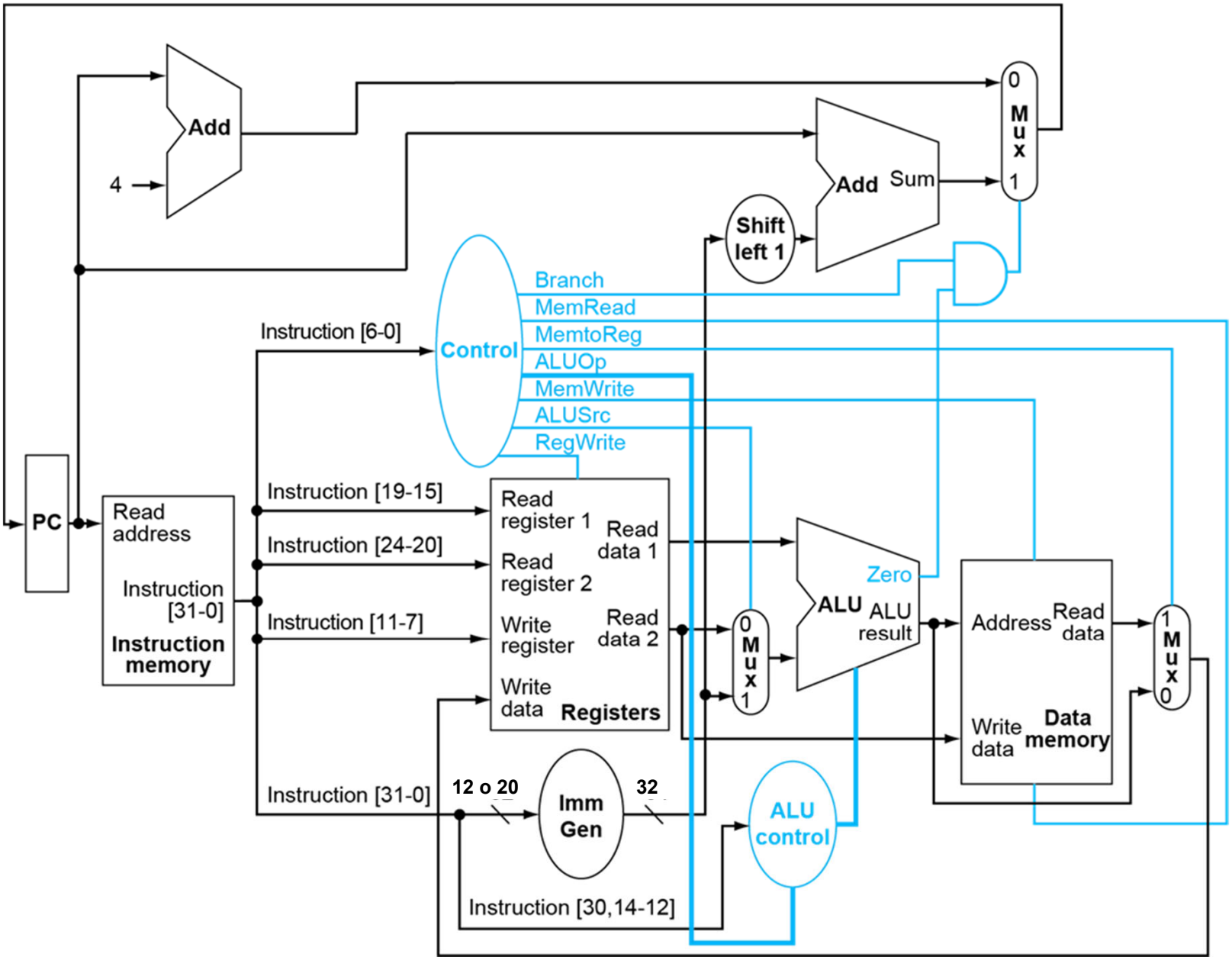
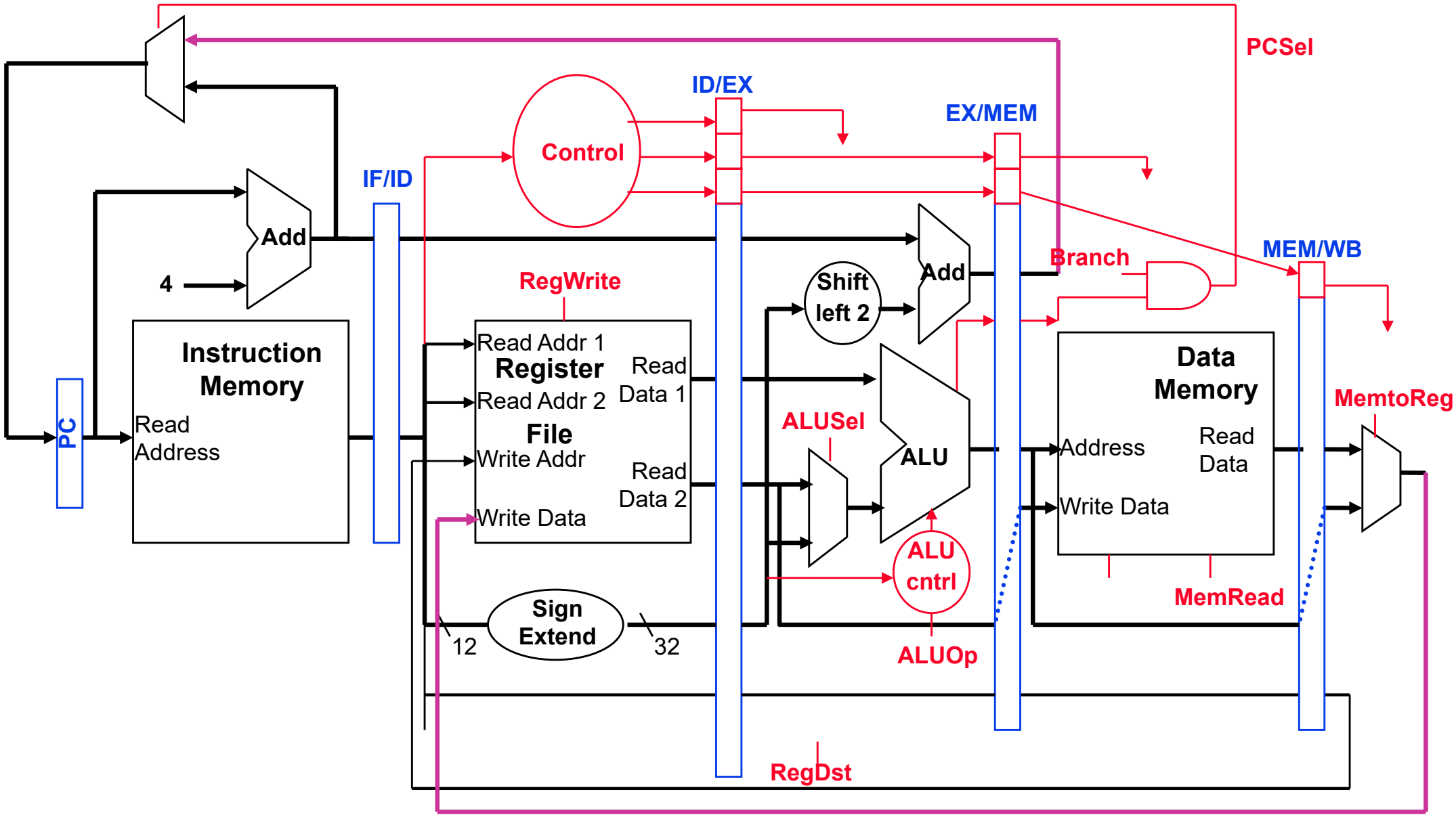
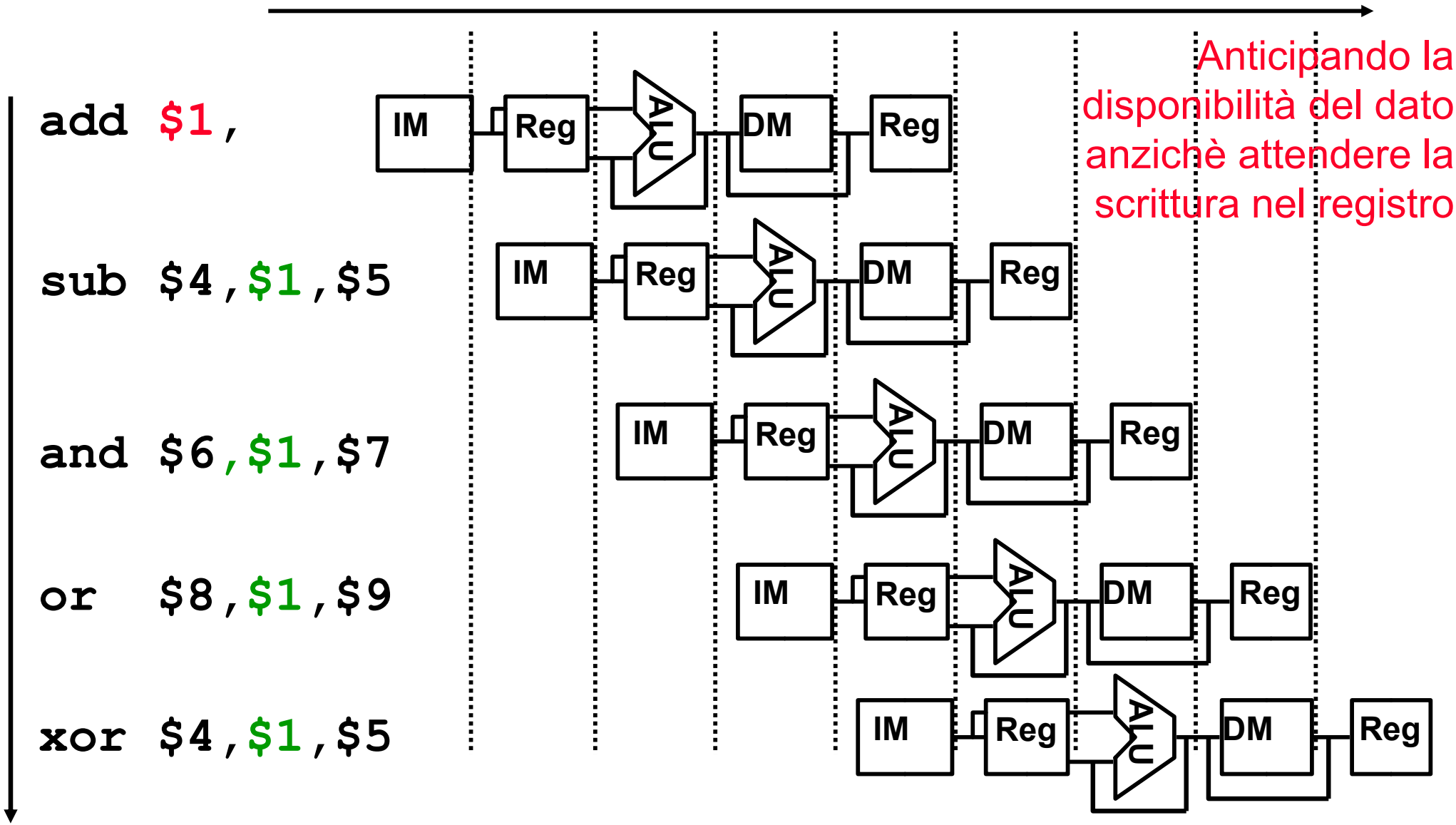
# RISC-V ciclo di clock unico



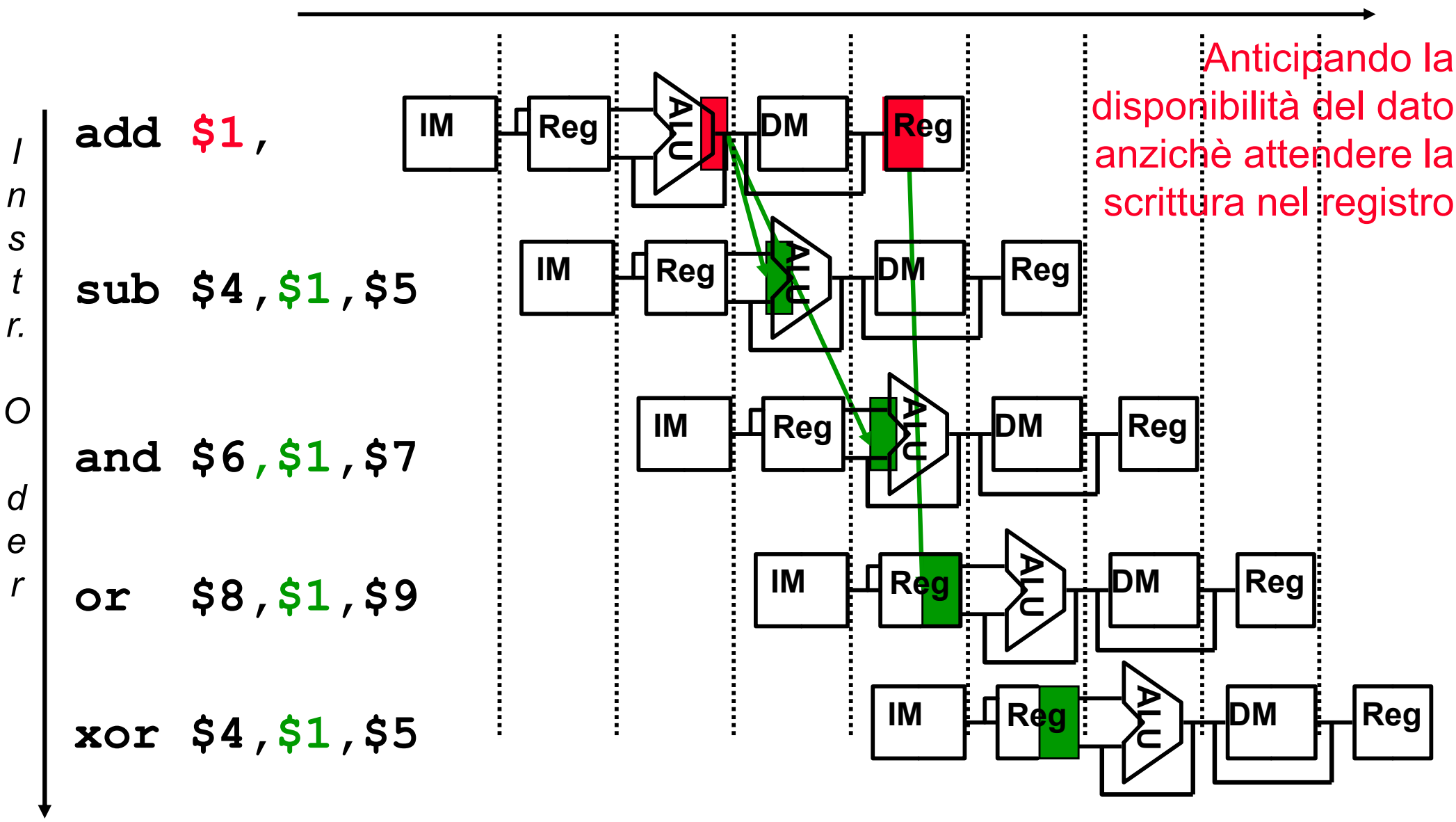
# Pipeline Control Path



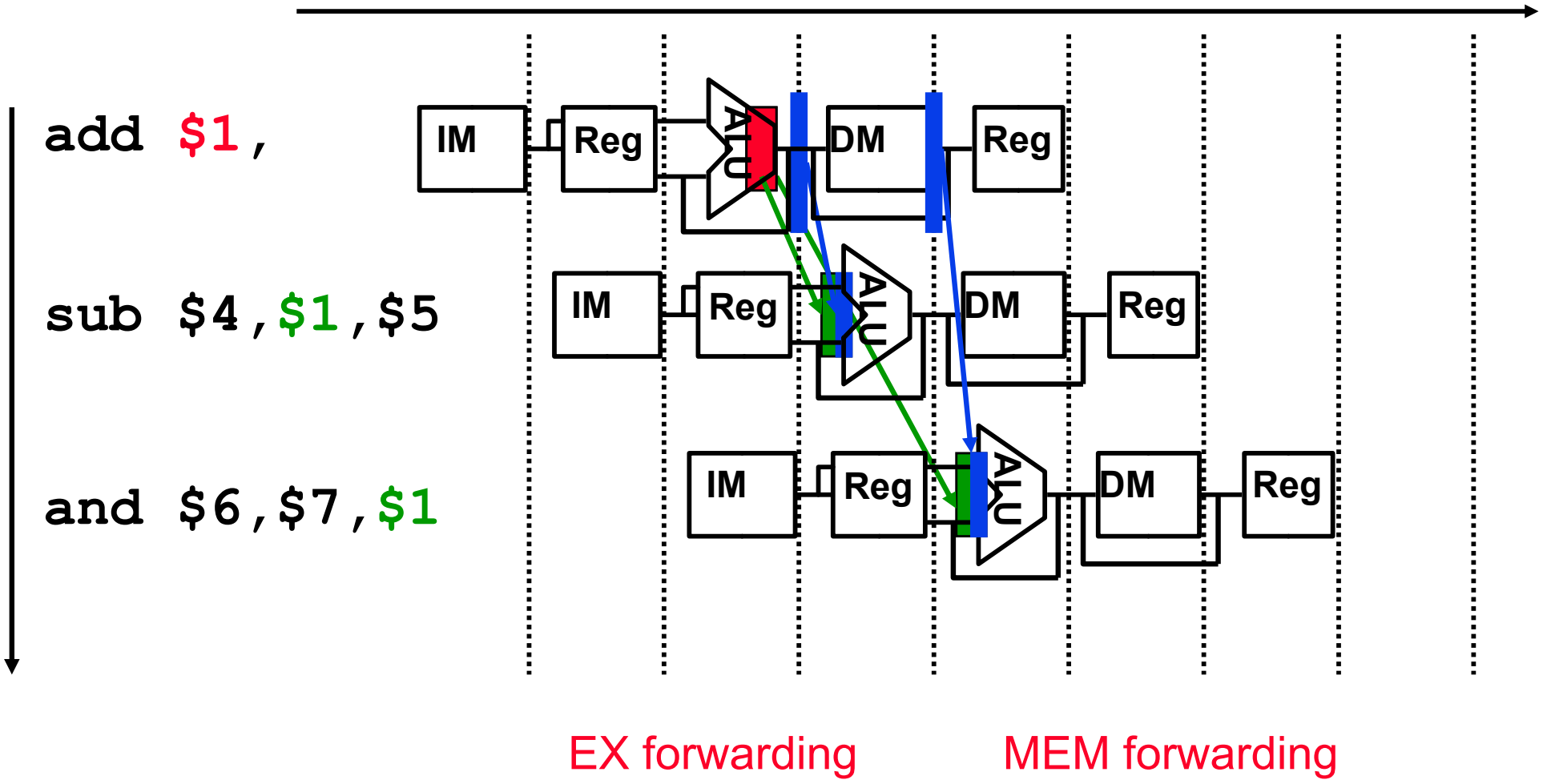
# Come evitare Data hazard



# Come evitare Data Hazard



# Forwarding


add $1,
sub $4,$1,$5
and $6,$7,$1
IM
Reg
ALU
DM
Reg
IM
Reg
ALU
DM
Reg
IM
Reg
ALU
DM
Reg
EX forwarding
MEM forwarding

# Data Forwarding : segnali di controllo

1. EX Forward Unit:

```
if (EX/MEM.RegWrite
and (EX/MEM.RegisterRd != 0)
and (EX/MEM.RegisterRd = ID/EX.RegisterRs))
      ForwardA = 10
if (EX/MEM.RegWrite
and (EX/MEM.RegisterRd != 0)
and (EX/MEM.RegisterRd = ID/EX.RegisterRt))
      ForwardB = 10
```

Da uscita ALU a ingresso ALU

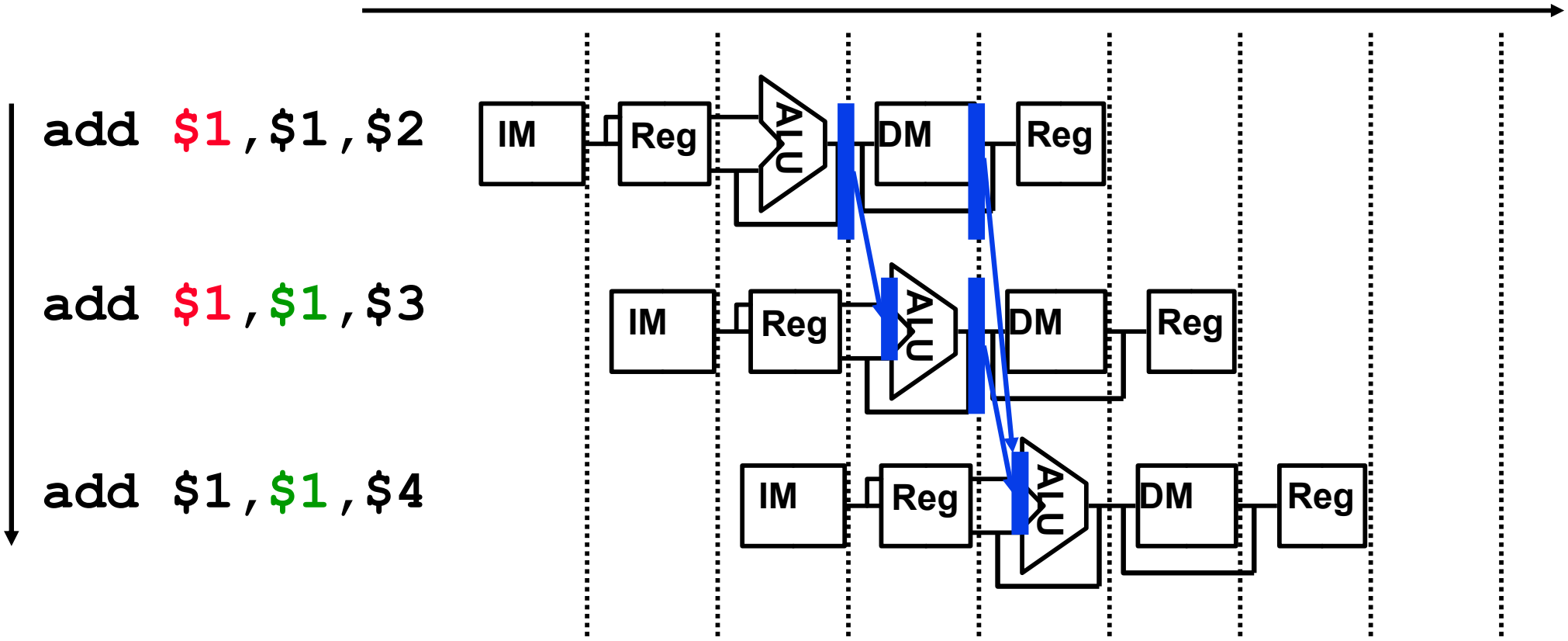
2. MEM Forward Unit:

```
if (MEM/WB.RegWrite
and (MEM/WB.RegisterRd != 0)
and (MEM/WB.RegisterRd = ID/EX.RegisterRs))
      ForwardA = 01
if (MEM/WB.RegWrite
and (MEM/WB.RegisterRd != 0)
and (MEM/WB.RegisterRd = ID/EX.RegisterRt))
      ForwardB = 01
```

Da uscita MEM a ingresso ALU

# Conflitto ripetuto

- Esempio: conflitto tra risultato dell'istruzione nella fase di WB con il risultato dell' istruzione in fase MEM



# Data Forwarding : segnali di controllo aggiornati

1. EX Forward Unit:

```
if (EX/MEM.RegWrite
and (EX/MEM.RegisterRd != 0)
and (EX/MEM.RegisterRd = ID/EX.RegisterRs))
      ForwardA = 10
if (EX/MEM.RegWrite
and (EX/MEM.RegisterRd != 0)
and (EX/MEM.RegisterRd = ID/EX.RegisterRt))
      ForwardB = 10
```

Da uscita ALU ad ingresso ALU

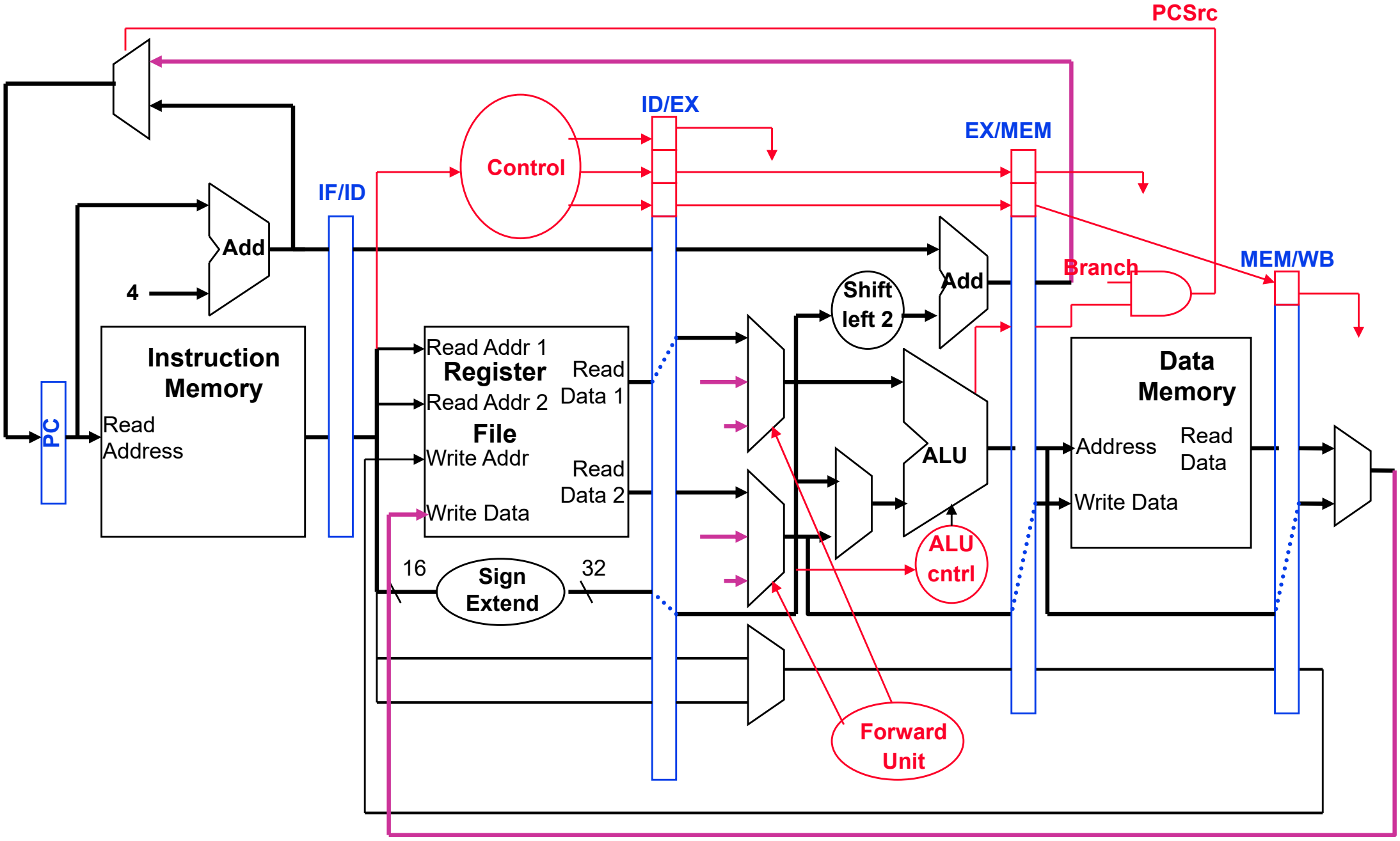
2. MEM Forward Unit:

```
if (MEM/WB.RegWrite
and (MEM/WB.RegisterRd != 0)
and (EX/MEM.RegisterRd != ID/EX.RegisterRs)
and (MEM/WB.RegisterRd = ID/EX.RegisterRs))
      ForwardA = 01

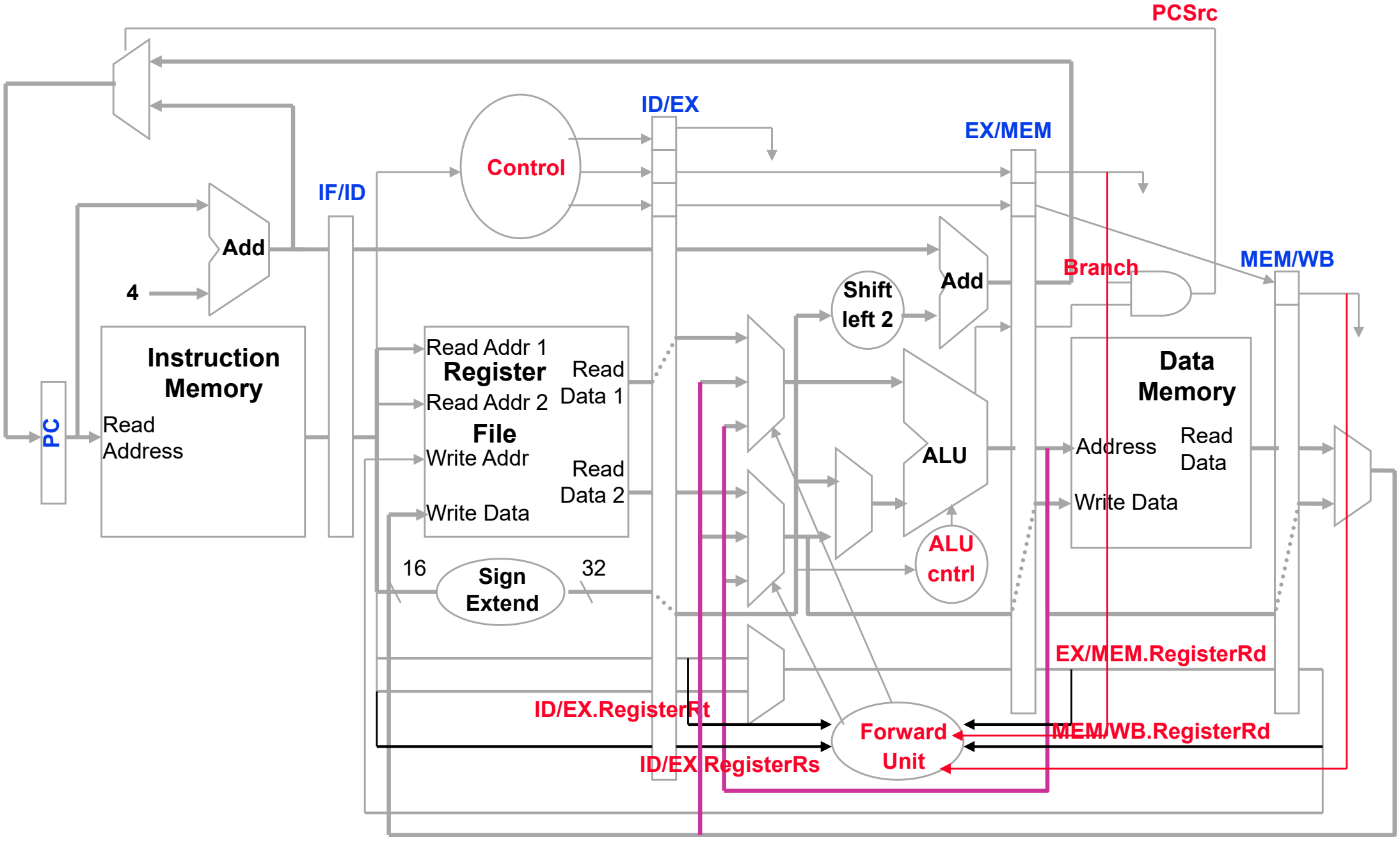
if (MEM/WB.RegWrite
and (MEM/WB.RegisterRd != 0)
and (EX/MEM.RegisterRd != ID/EX.RegisterRt)
and (MEM/WB.RegisterRd = ID/EX.RegisterRt))
      ForwardB = 01
```

Da uscita ALU oppure MEM ad ingresso ALU

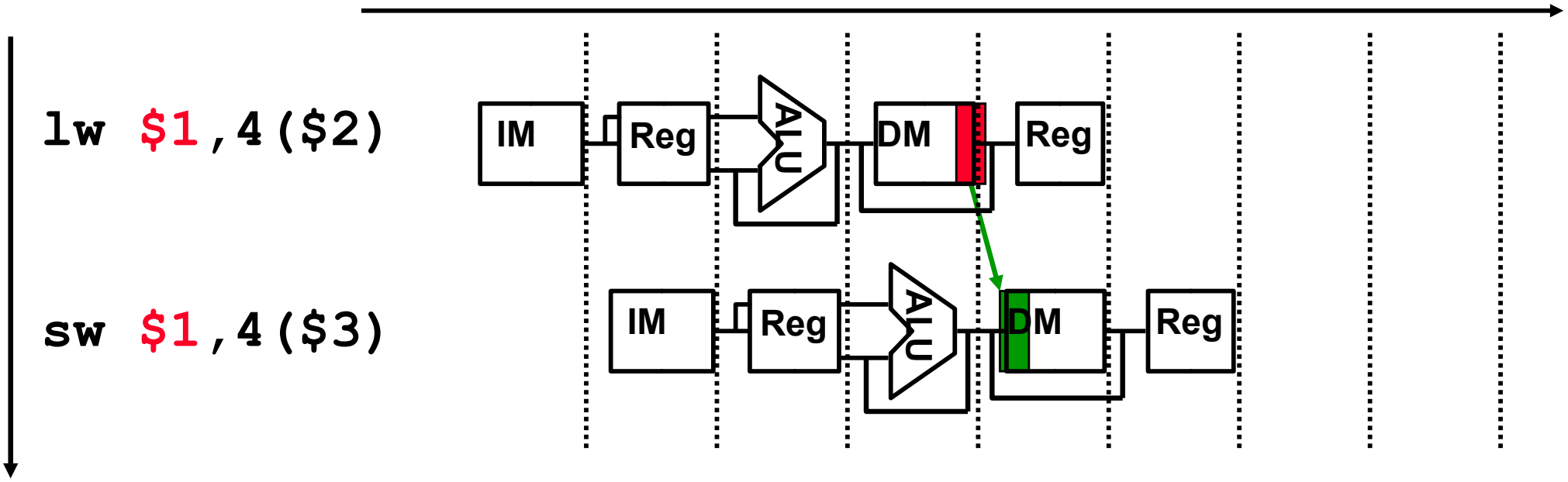
# Datapath con forwarding



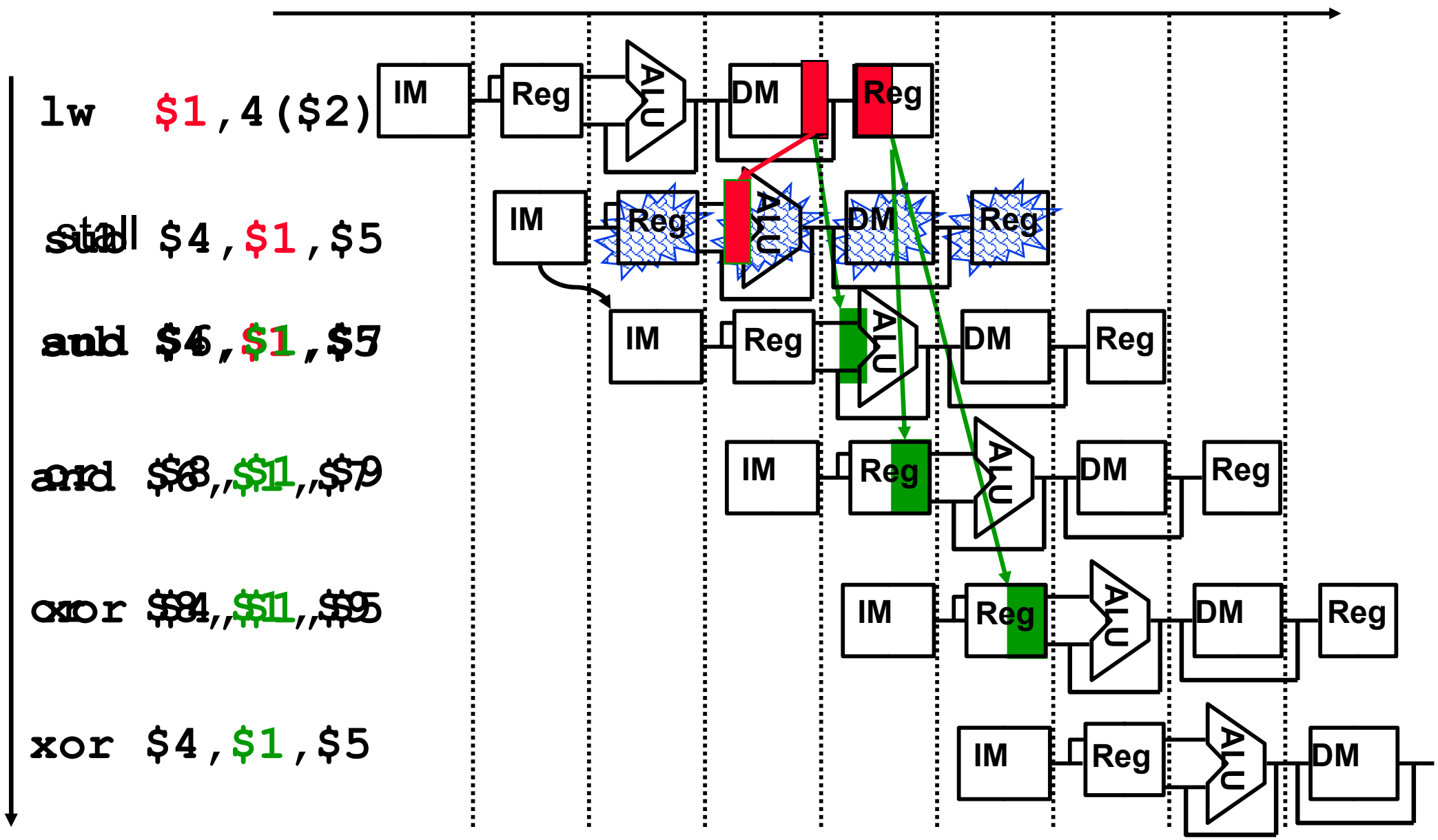
# Datapath con Forwarding



# Copia dati from-Memory-to-Memory



# Conflitti di tipo load-use

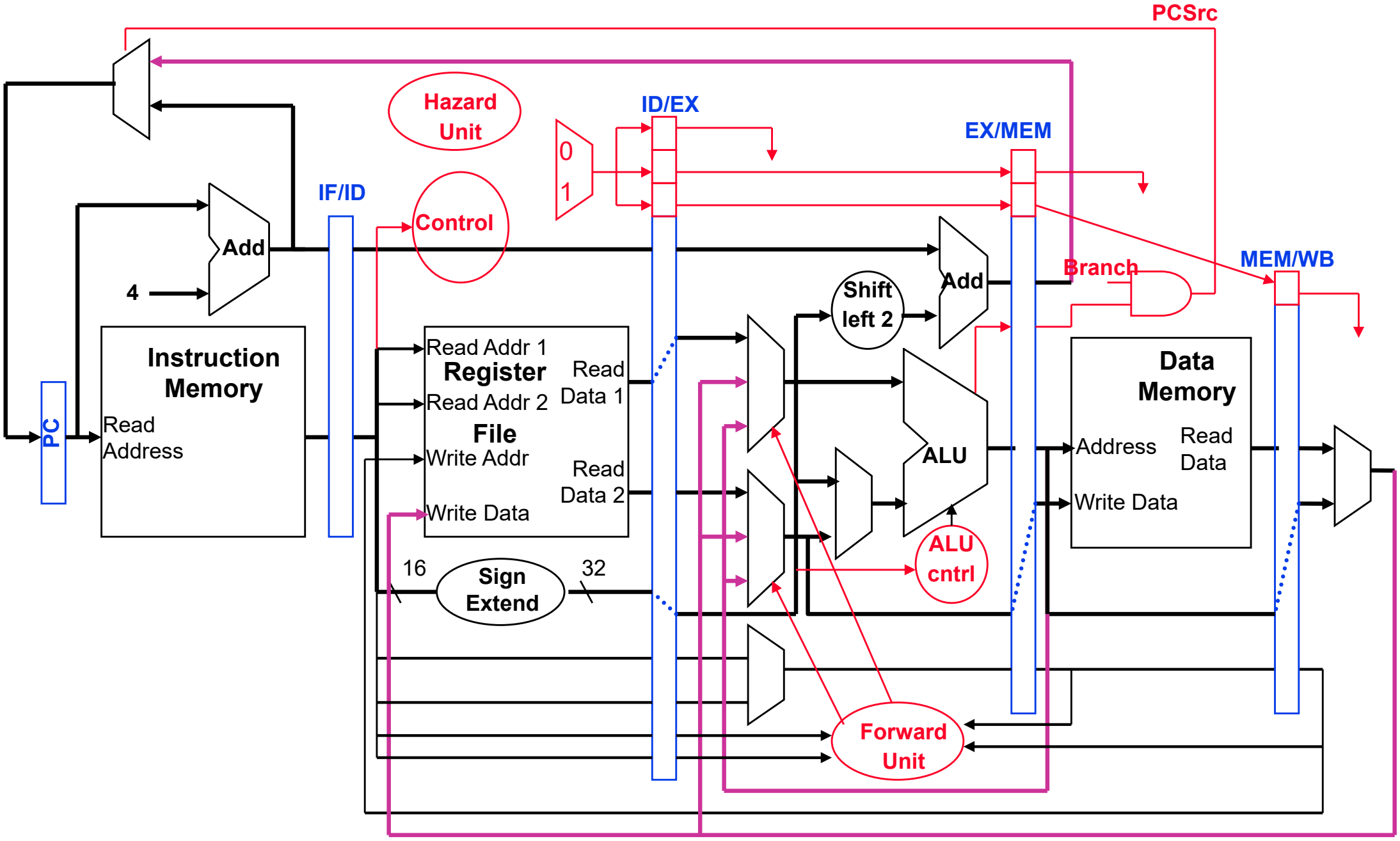


# Riconoscimento di conflitti Load-use

1. ID Hazard detection Unit:

```
if (ID/EX.MemRead
and ((ID/EX.RegisterRt = IF/ID.RegisterRs)
or  (ID/EX.RegisterRt = IF/ID.RegisterRt)))
stall the pipeline
```

# Stallo inserito in HW



# Stallo inserito in HW

